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Schuster et al.

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(54) **SPEAKER WITH BATCHED CONES**

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H04R 1/02 (2006.01)

(52) **U.S. Cl.**
CPC ***H04R 1/025*** (2013.01); *H04R 2201/02* (2013.01); *H04R 2400/11* (2013.01)

(58) **Field of Classification Search**
CPC H04R 1/025; H04R 2201/02; H04R 2400/11; H04R 9/043; H04R 9/063; H04R 9/06; H04R 9/02
See application file for complete search history.

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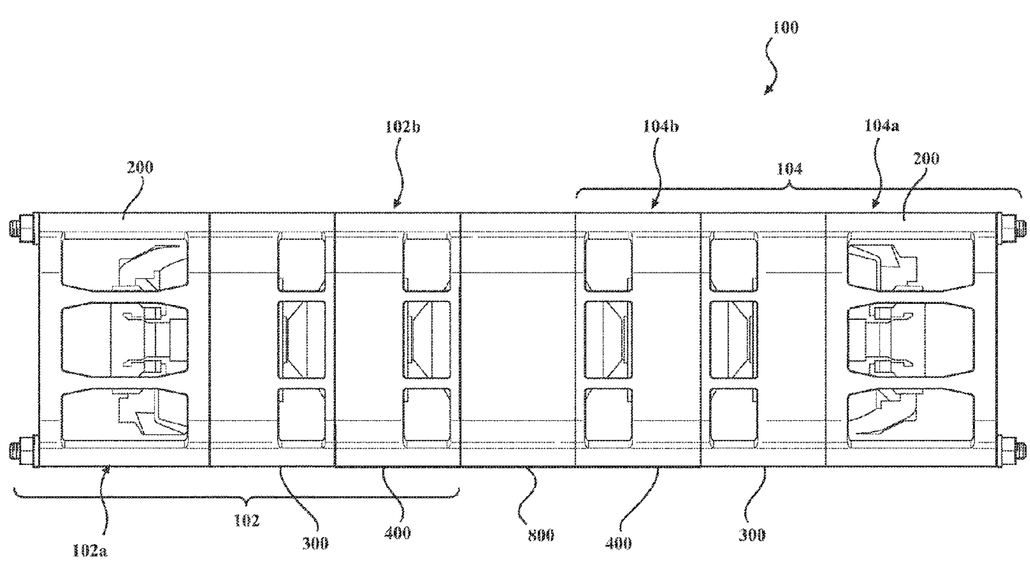
(57) **ABSTRACT**

A speaker sub-assembly has at least one active segment base segment, at least one batch segment, and a passive end segment. In one or more embodiments, speaker sub-assemblies with batched cones may be attached to each other by way of an adapter segment to create a dual-core speaker with batched cones. In one or more embodiments, speaker sub-assemblies with batched cones may be assembled in parallel, or linearly end-to-end.

9 Claims, 13 Drawing Sheets



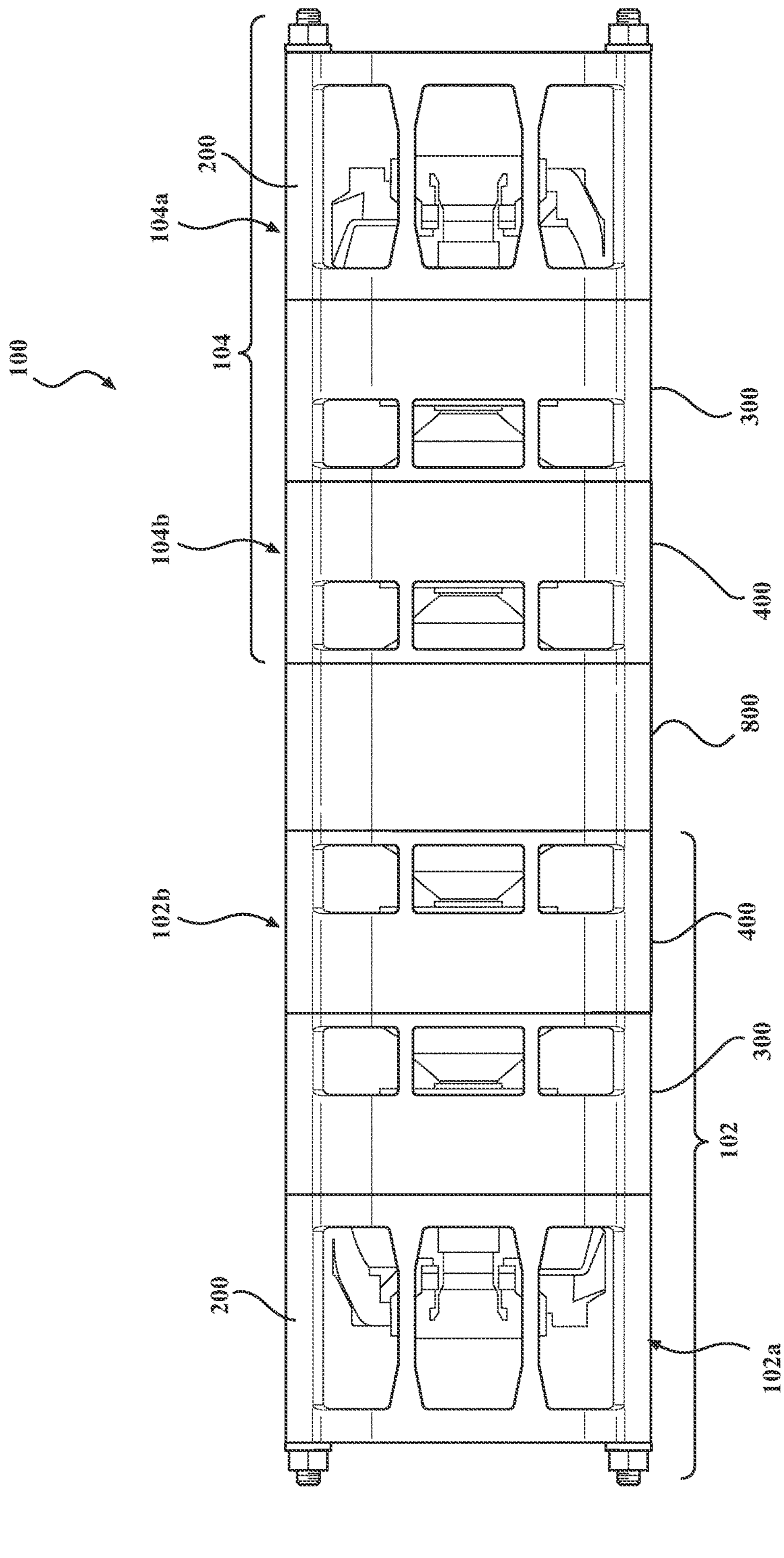
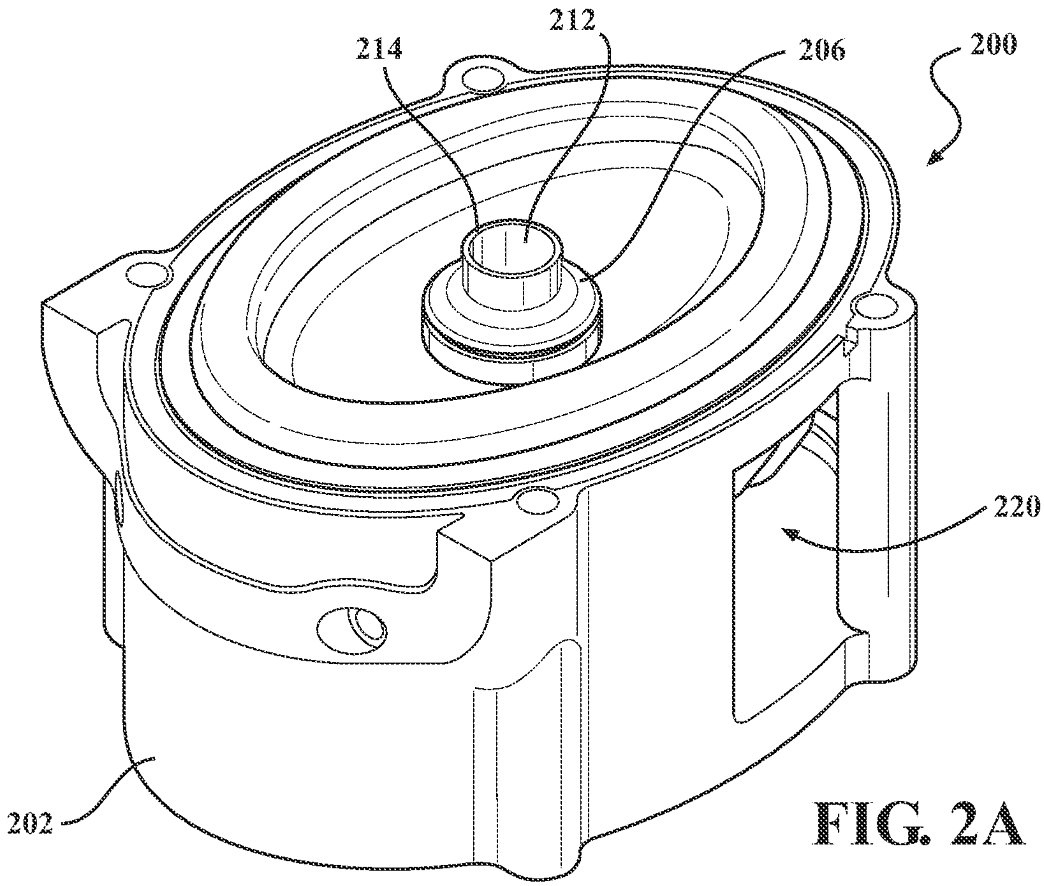
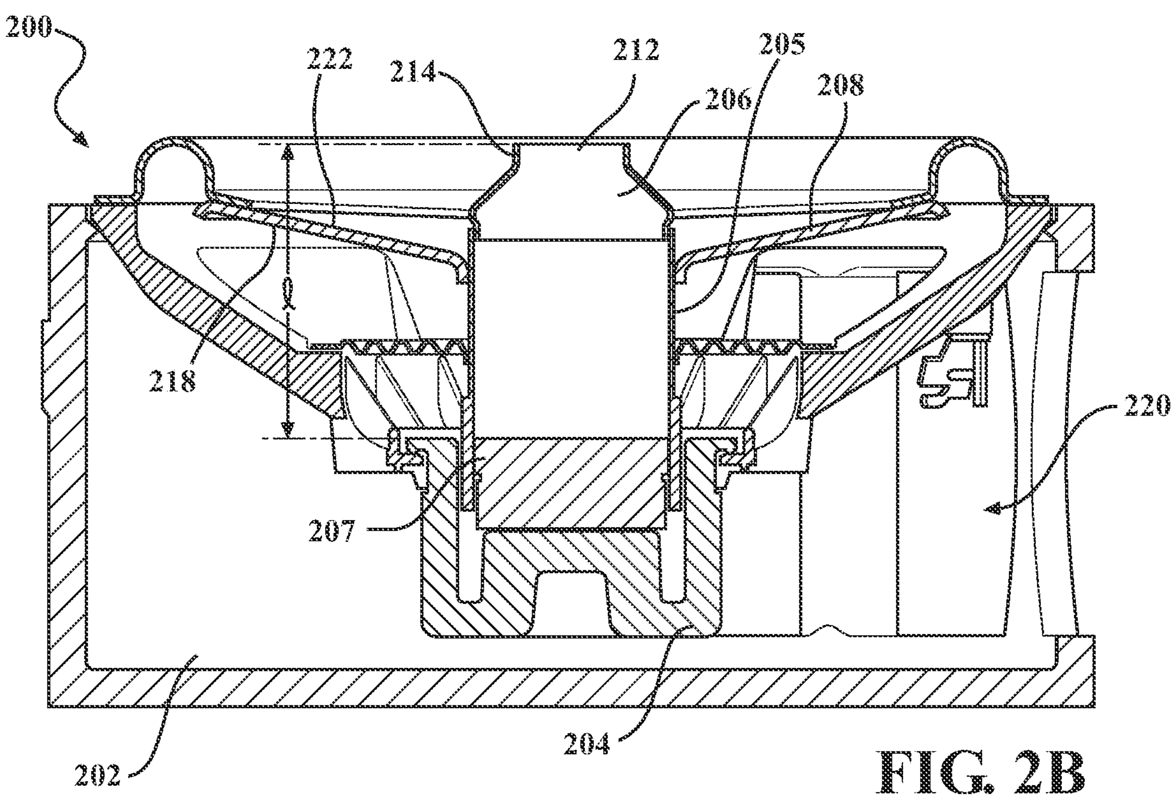


FIG. 1



212
214
206
200
220
202

FIG. 2A



200
222
214
212
205
206
208
218
220
207
202
204

FIG. 2B

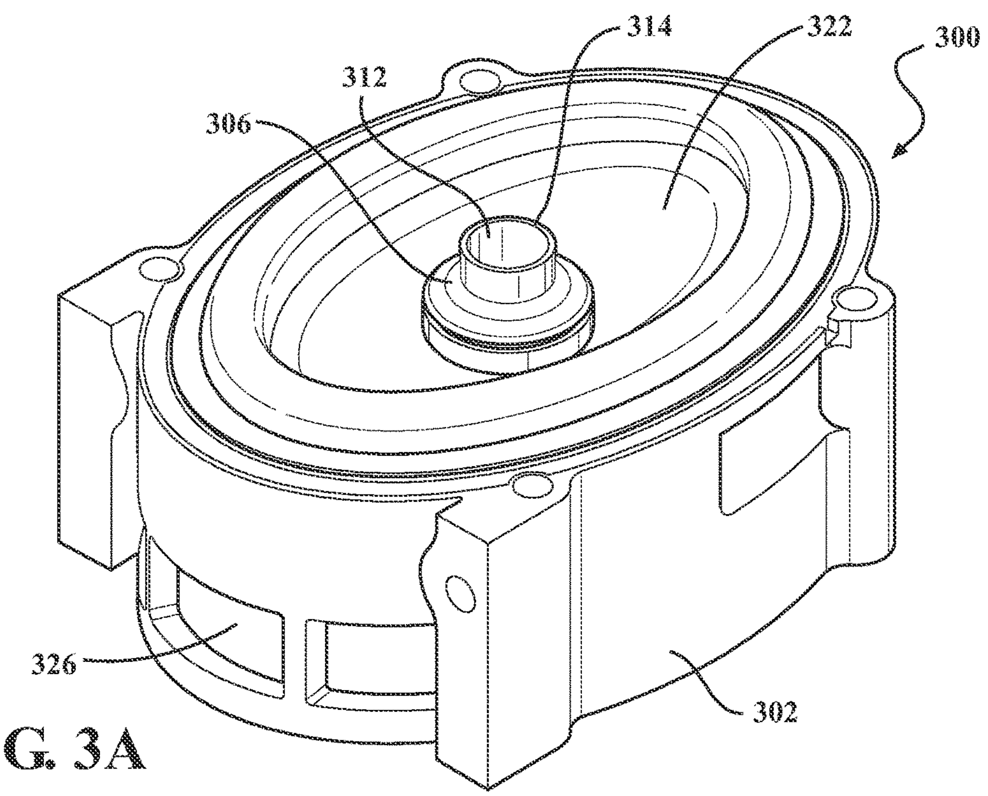

314
322
300
312
306
326
302

FIG. 3A

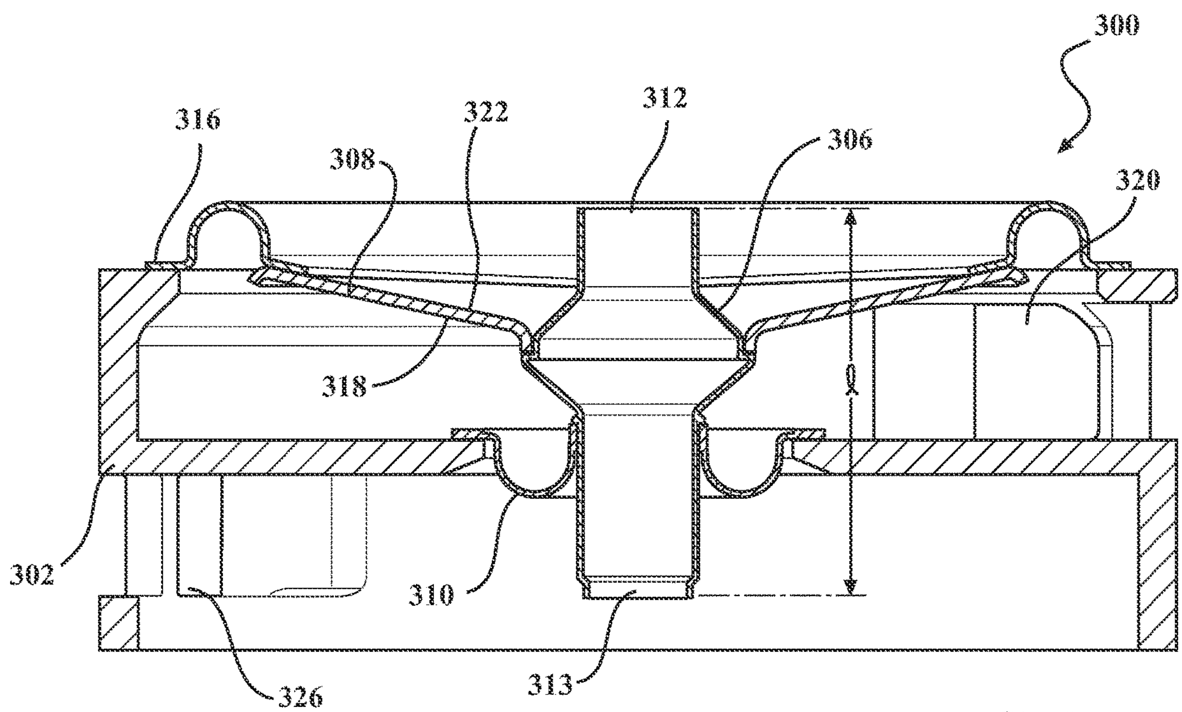

300
316
322
312
306
308
320
318
302
310
313
326

FIG. 3B

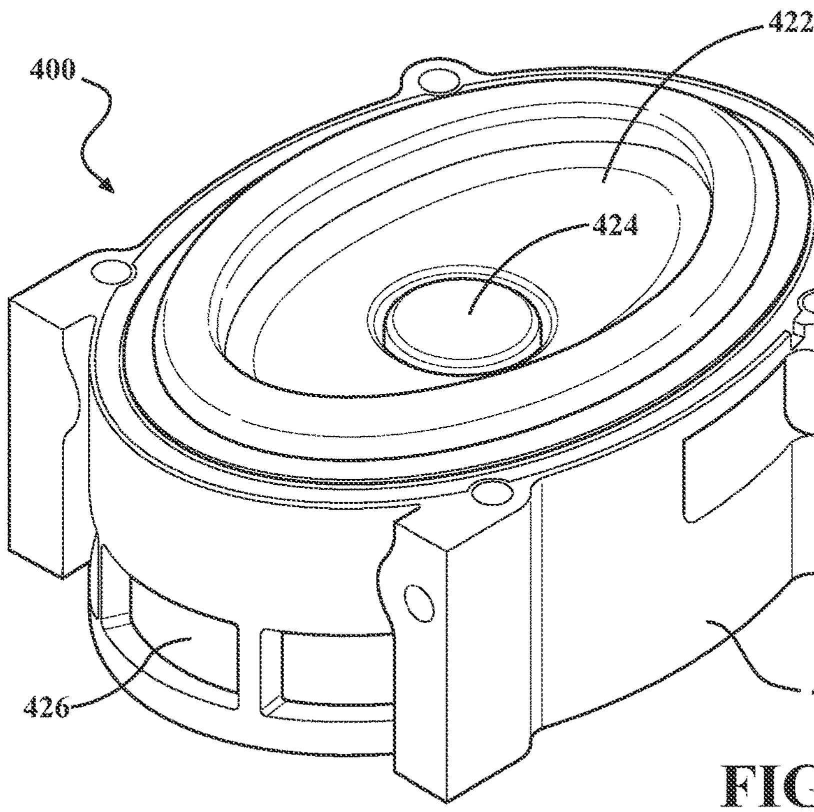

400
422
424
426
402

FIG. 4A

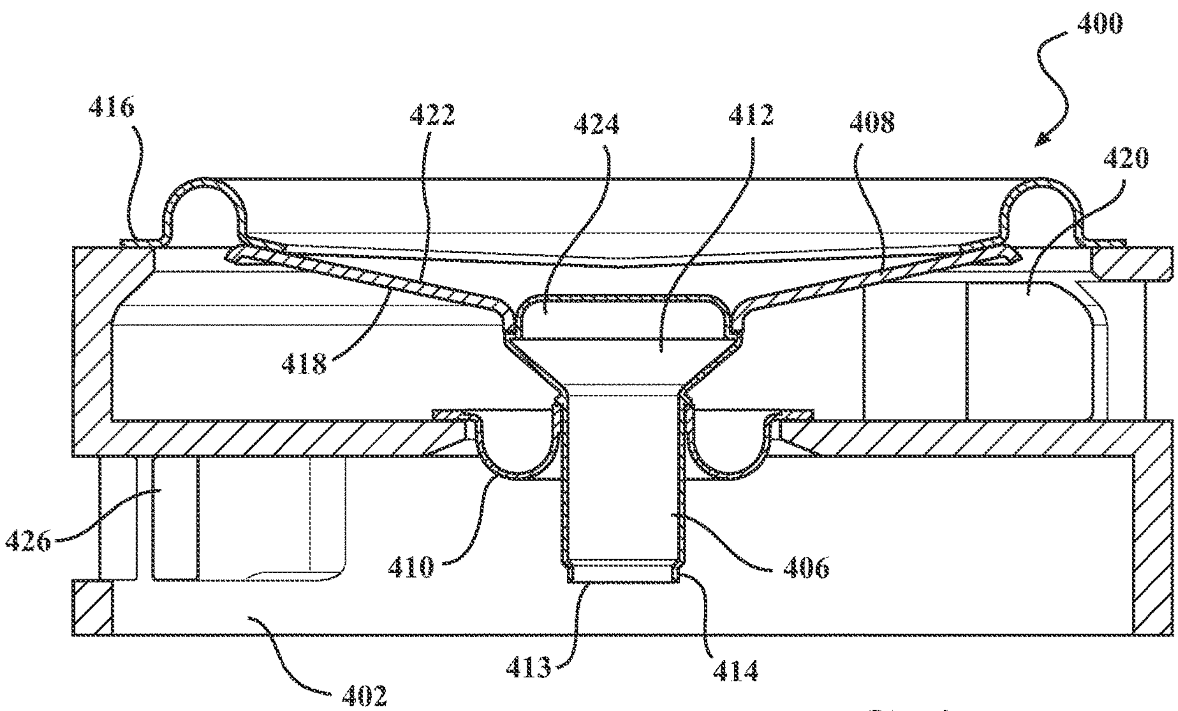

400
416
422
424
412
408
420
418
426
410
406
413
414
402

FIG. 4B

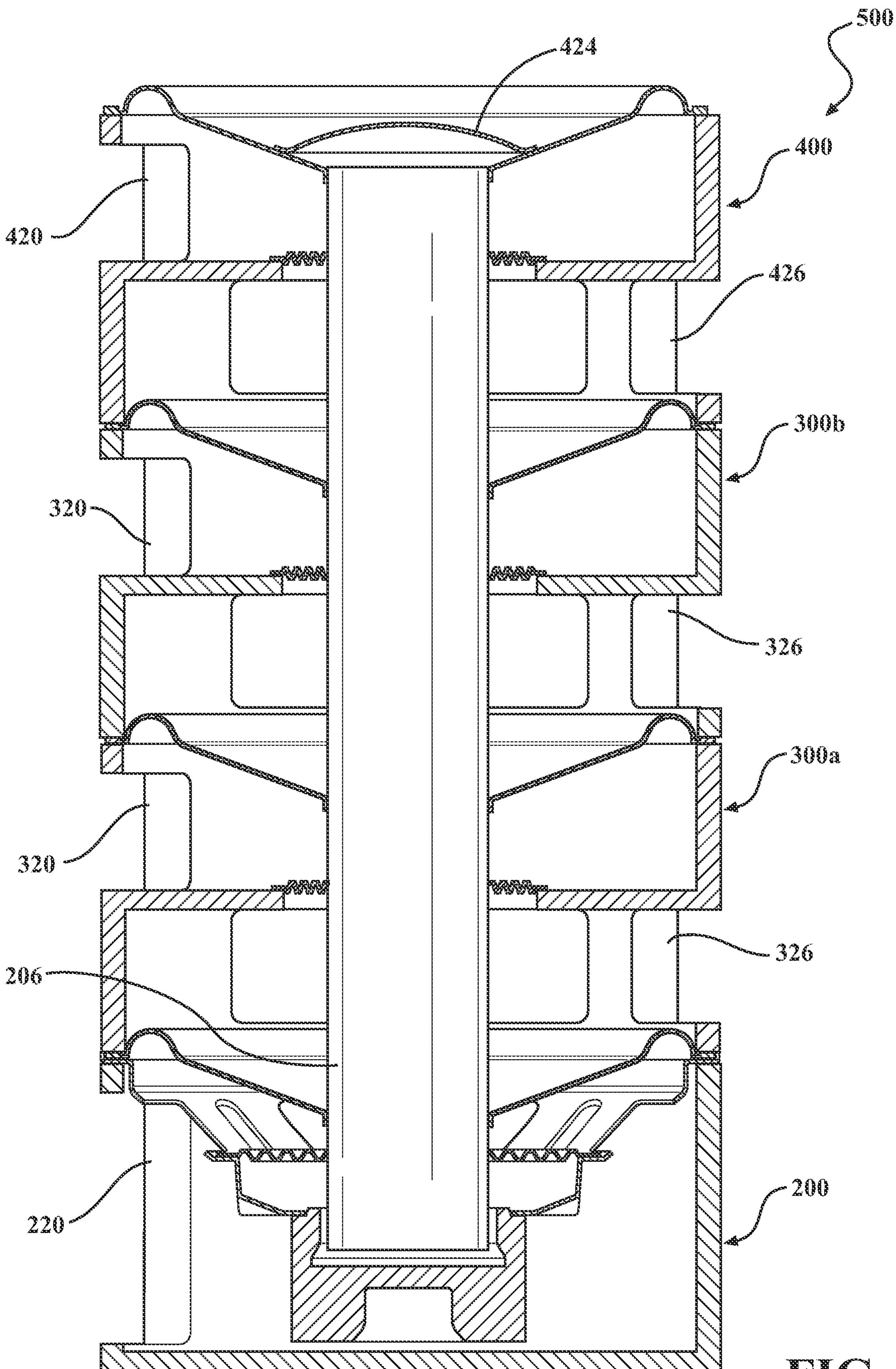


FIG. 5

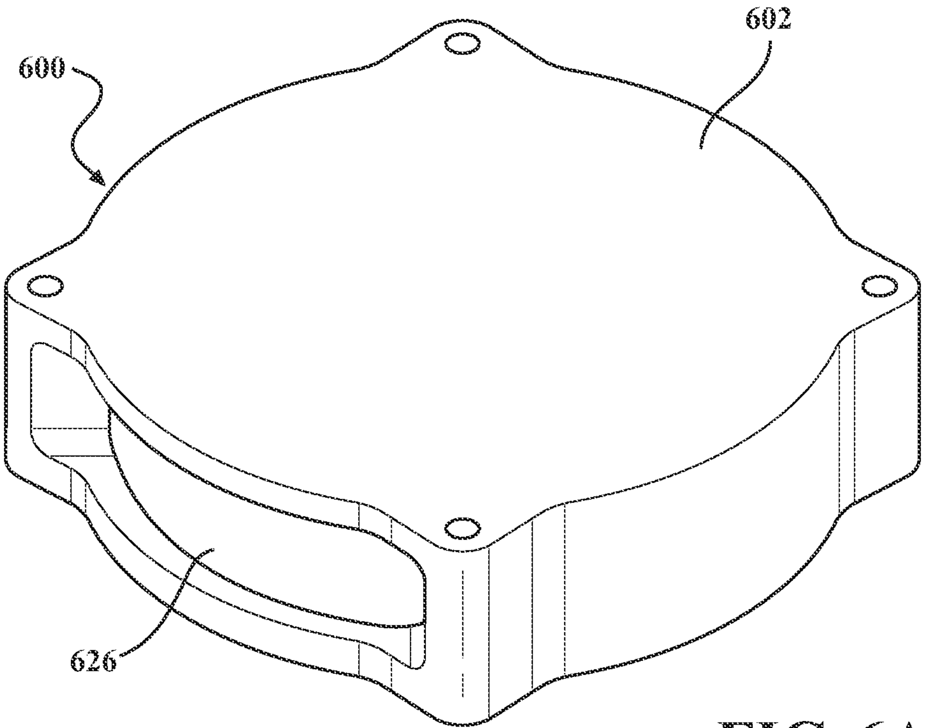


FIG. 6A

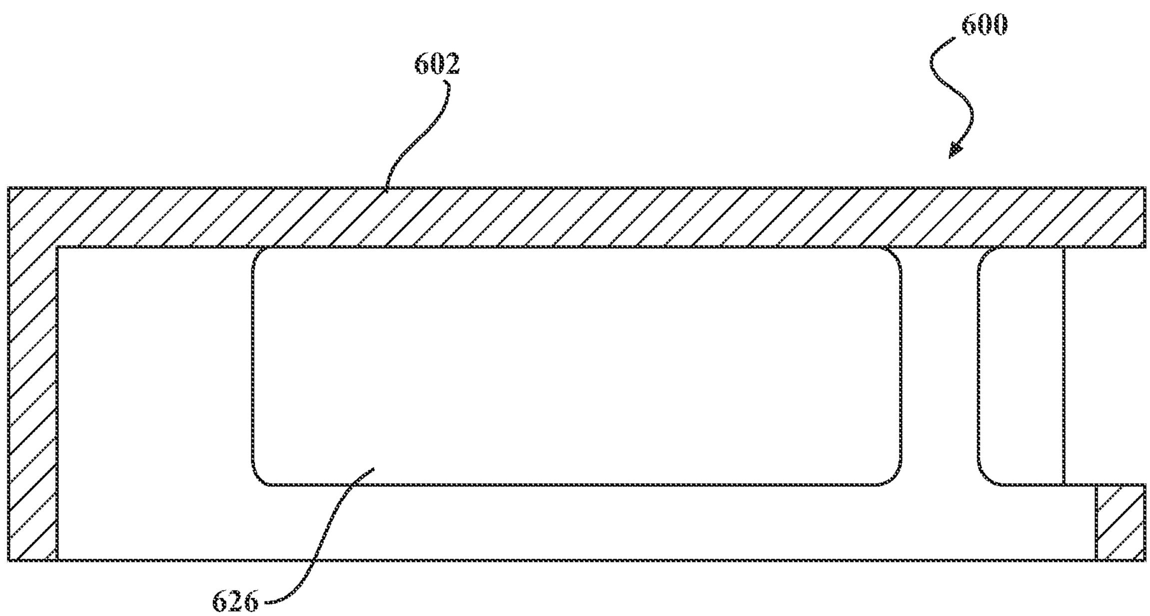


FIG. 6B

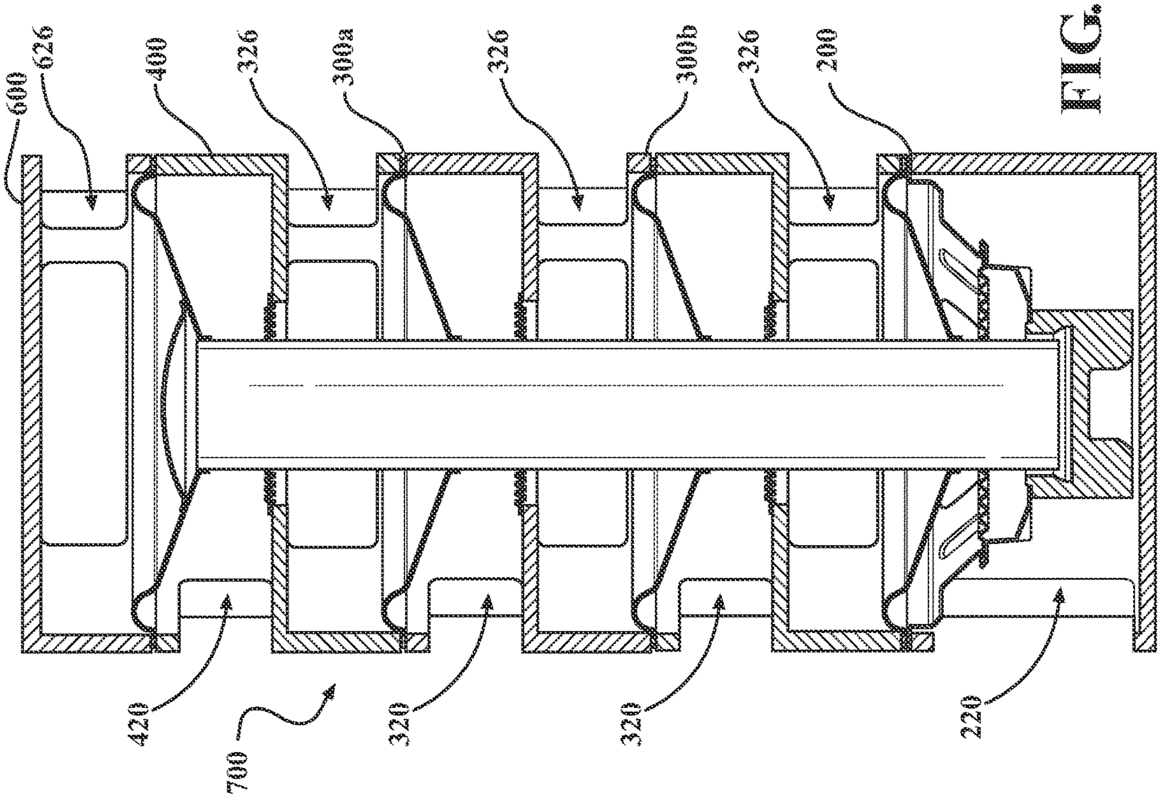

600
626
400
326
300a
326
300b
326
200
420
700
320
320
220

FIG. 7B

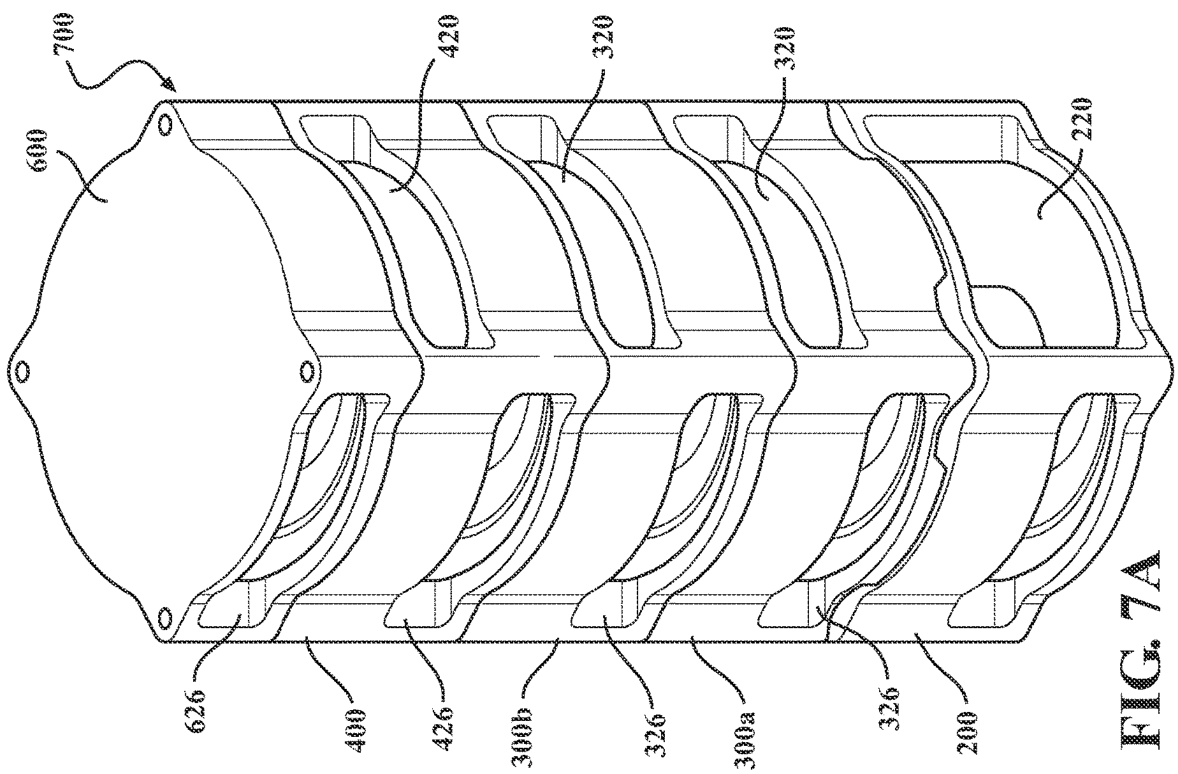

700
600
420
320
320
220
626
400
426
300b
326
300a
326
200

FIG. 7A

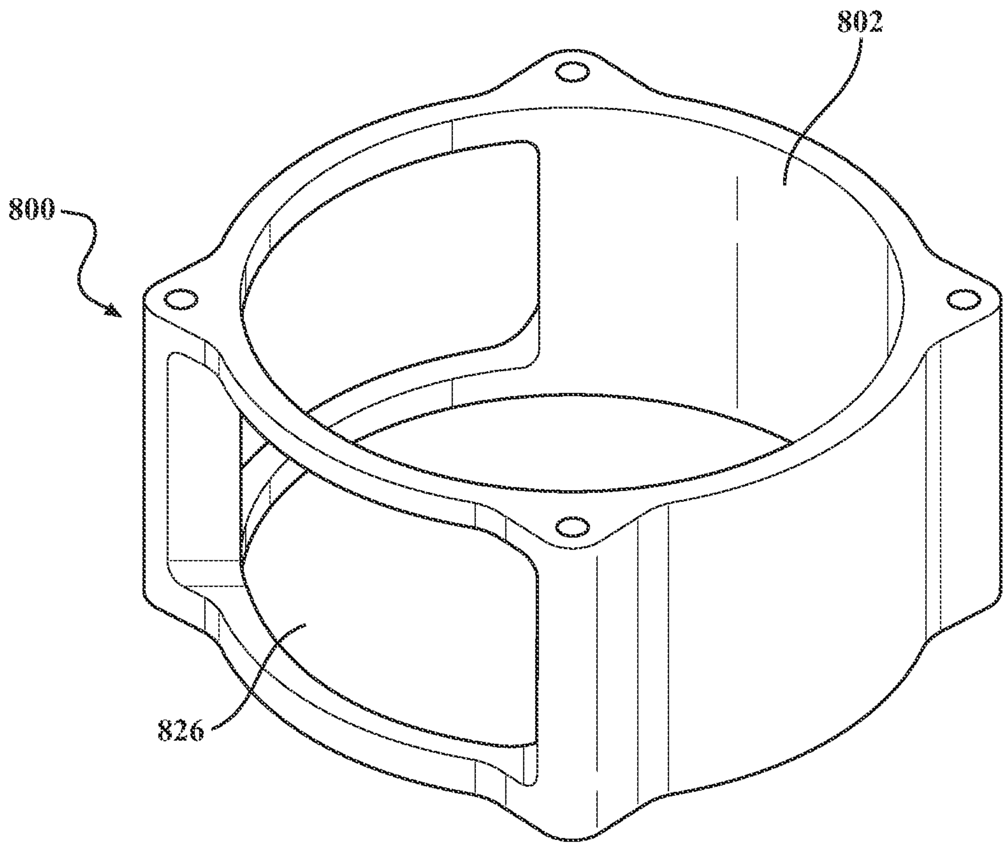


FIG. 8A

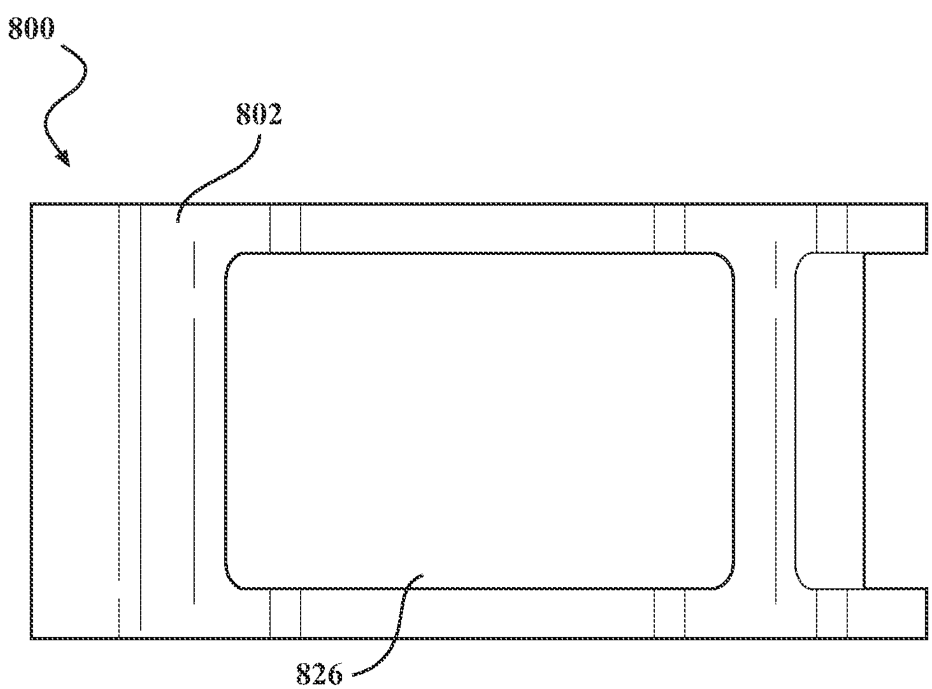


FIG. 8B

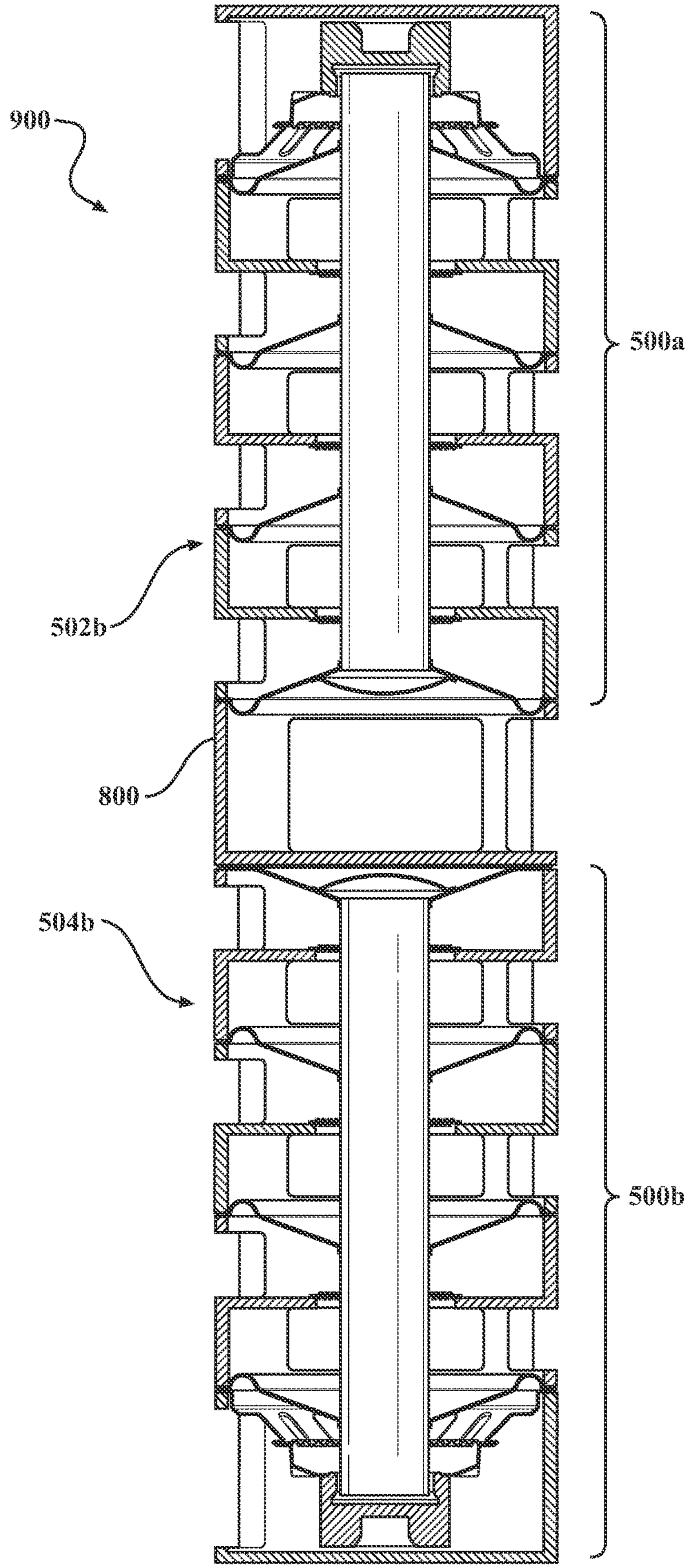


FIG. 9

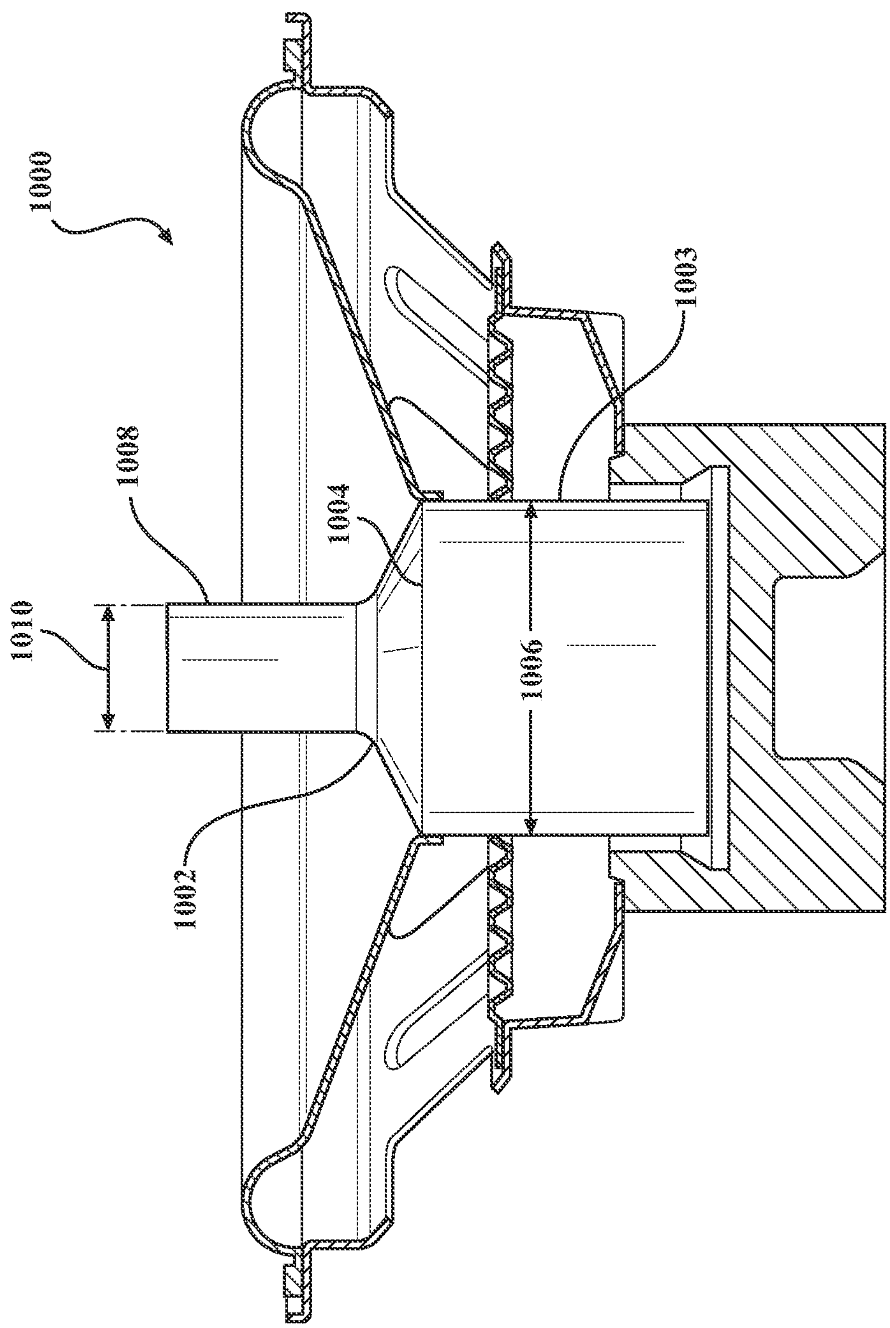


FIG. 10

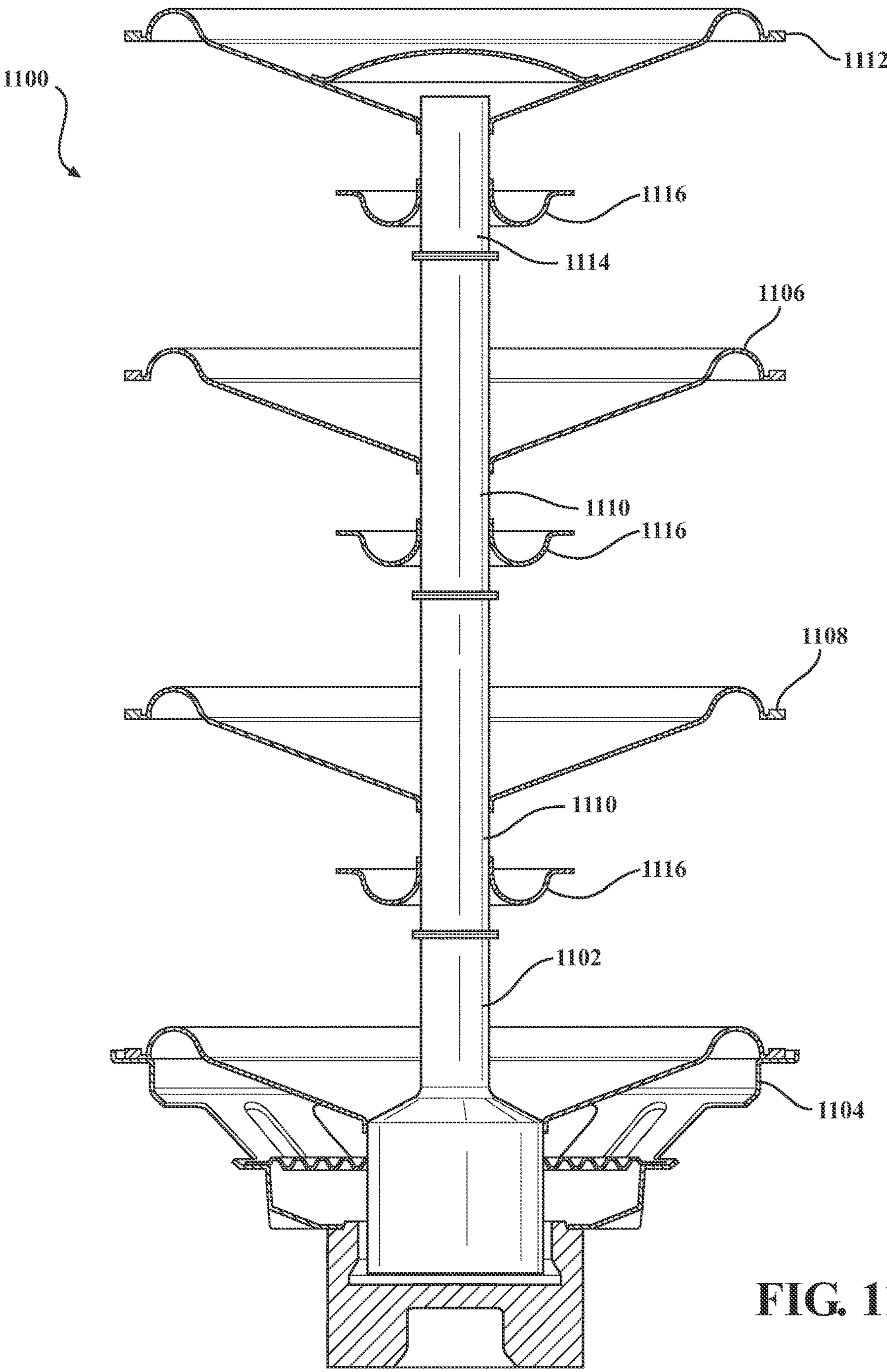


FIG. 11

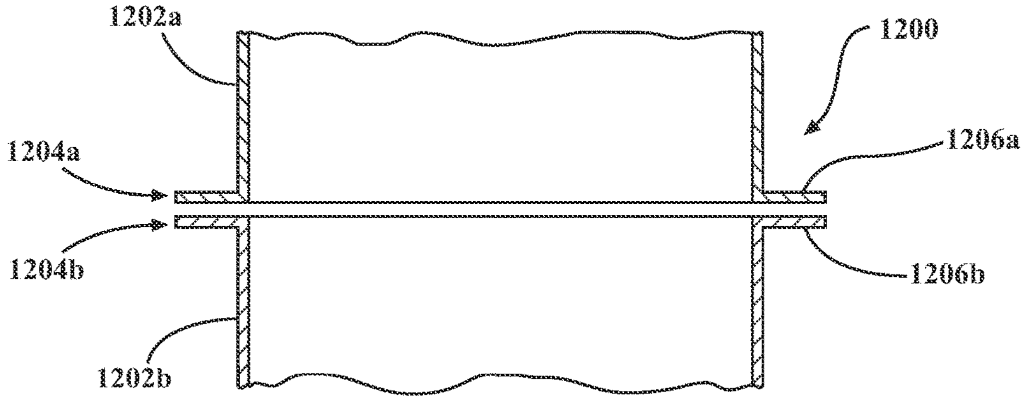


FIG. 12

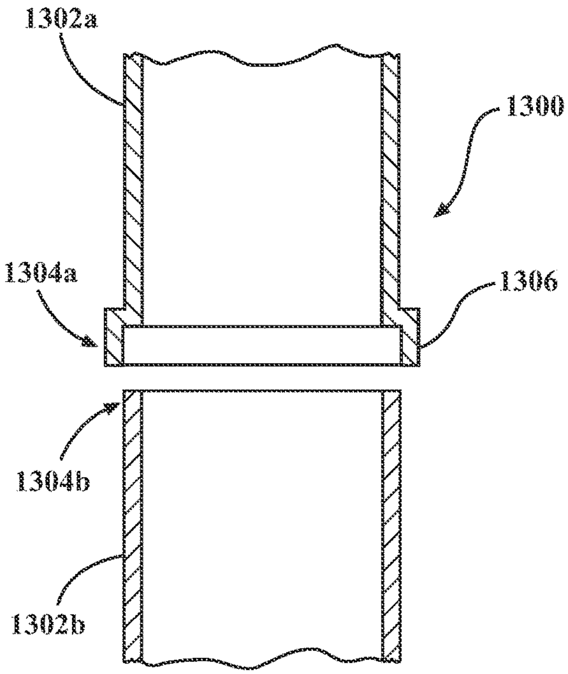


FIG. 13

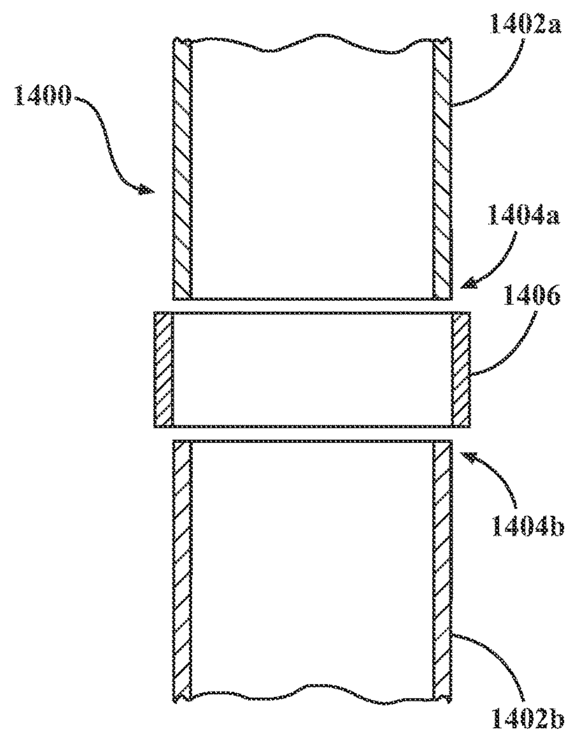


FIG. 14

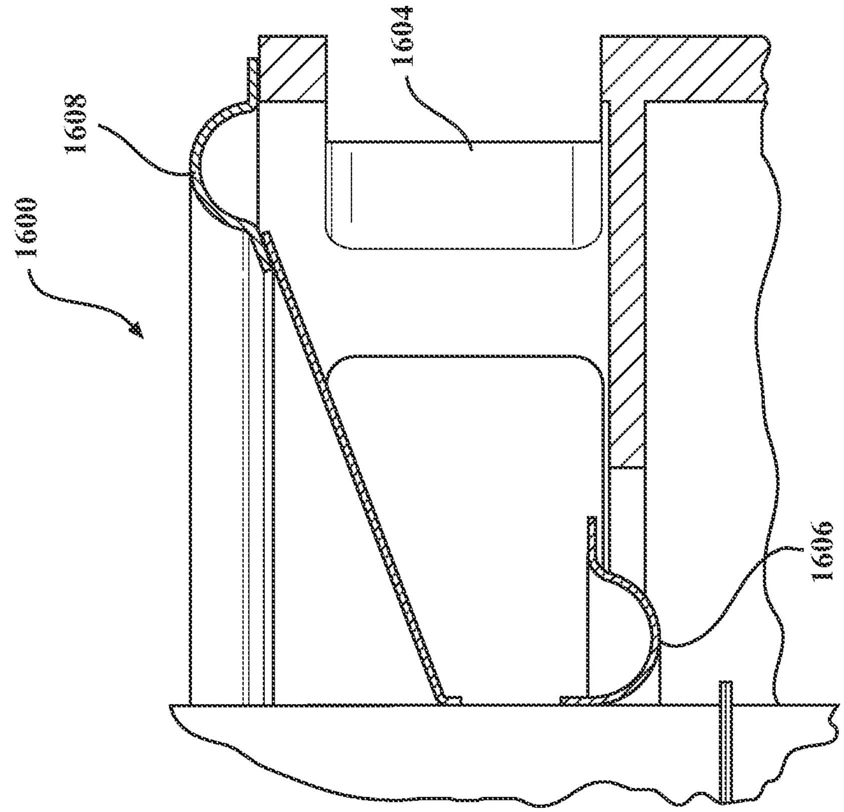


FIG. 16

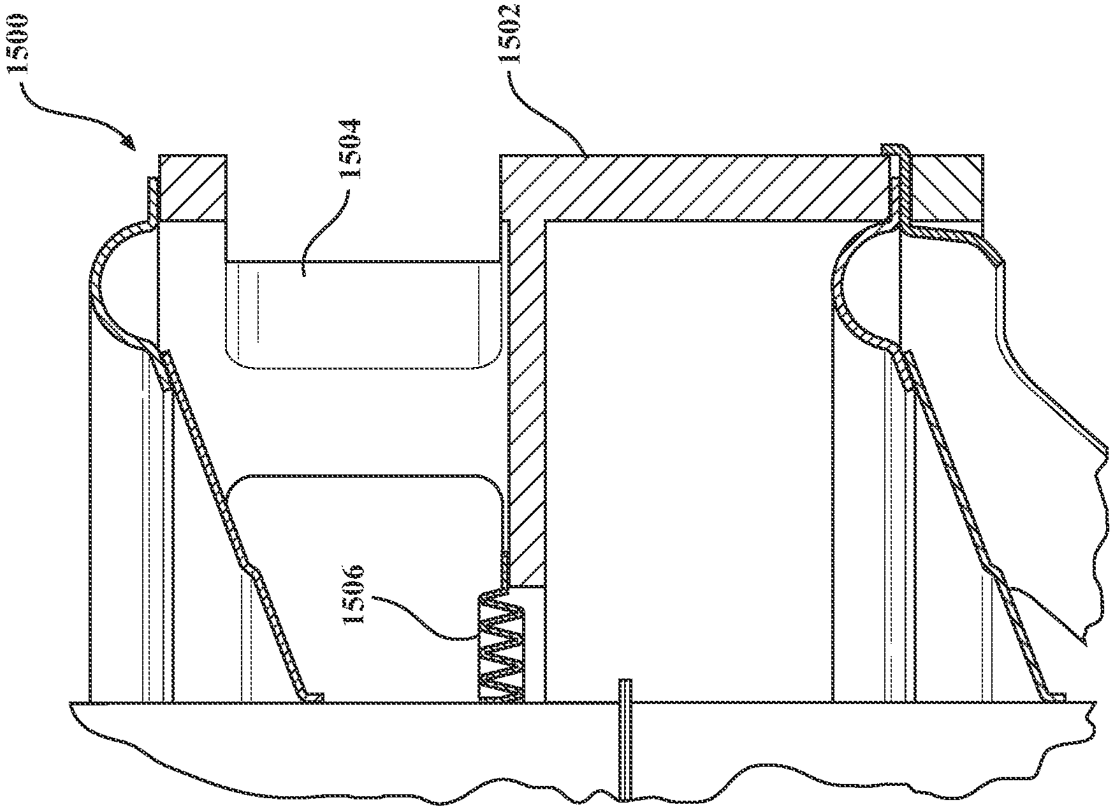


FIG. 15

SPEAKER WITH BATCHED CONES

TECHNICAL FIELD

The present disclosure relates to a speaker and more particularly, to a speaker with batched cones.

BACKGROUND

Audio systems in a listening environment typically include speakers that are distributed around the vehicle interior. Premium audio systems often include large subwoofers and dedicated amplifiers. In general, a premium audio system requires a lot of available space in the listening environment and locations within the available space that speakers may be positioned. Configuring an audio system involves selecting speaker locations in the listening environment. For a listening environment with limited space, such as a vehicle cabin, locating speakers, especially low frequency operating speakers such as subwoofers, packaging constraints are challenging. Generation of low frequency needs a large cabin volume and transducer size. Therefore, in automotive scenarios a compromise in cabinet size and in transducer size must be made, which usually limits low-frequency cutoff of the audio system.

Furthermore, a smaller listening environment requires smaller speakers, which inherently are not as efficient in generating midrange and low frequencies as larger speakers. A listener has an expectation for bass performance that may be adversely affected by smaller speakers. Solutions such as psychoacoustic tuning attempt to compensate for reduced bass-performance, but at the expense of clarity and high-fidelity performance.

Another solution is to attempt a more compact speaker design by including passive radiators or using a vented enclosure. However, these solutions have drawbacks of their own. Passive radiators often cause the speaker to move noticeably resulting in unwanted noise. Vented enclosures may expose the speaker to harsh environments. Furthermore, such speaker designs are still subject to the size/location limitations associated with the listening environment, thereby requiring their design be on a case-by-case basis.

There is a need for a transducer assembly in a speaker that, unlike a conventional speaker, is capable of scalable variations in size to accommodate a variety of packaging conditions while maintaining performance.

SUMMARY

A speaker sub-assembly having at least one active segment base segment, at least one batch segment, and a passive end segment. In one or more embodiments, speaker sub-assemblies with batched cones may be attached to each other by way of an adapter segment to create a dual-core speaker with batched cones. In one or more embodiments, speaker sub-assemblies with batched cones may be assembled in parallel, or linearly end-to-end.

In one or more embodiments, a scalable speaker assembly has at least one electrically driven base segment having a motor and a voice coil driven by the motor, a passive end segment having a cone driven by the at least one electrically driven base segment, and a plurality of passive batch segments connected between the at least one electrically driven base segment and the passive end segment, each batch segment in the plurality of batch segments is stacked to provide an overall cone area for the scalable speaker assembly. In one or more embodiments, two scalable speaker assemblies may be attached to each other by way of an adapter segment to create a dual-core speaker with batched cones. In one or more embodiments, a plurality of dual-core speakers may be assembled to each other in parallel, or linearly end-to-end.

DESCRIPTION OF DRAWINGS

FIG. **1**. is a front view of one or more embodiments having two scalable speakers with batched cones attached at a passive end segment of each speaker with batched cones;

FIG. **2**A is a perspective view of a base segment of a scalable speaker with batched cones;

FIG. **2**B is a cross-sectional view of the base segment;

FIG. **3**A is a perspective view of a batch segment;

FIG. **3**B is a cross-sectional view of the batch segment;

FIG. **4**A is a perspective view of an end segment;

FIG. **4** B is a cross-sectional view of the end segment;

FIG. **5** is a cutaway view of a base segment, two batch segments and an end segment forming a scalable speaker with batched cones;

FIG. **6**A is a perspective view of a cover segment;

FIG. **6**B is a cross-sectional view of the cover segment;

FIG. **7**A is a scalable speaker with batched cones having a base segment, two batch segments, an end segment, and a cover segment;

FIG. **7**B is a cross-sectional view of the scalable speaker with batched cones of FIG. **7**A;

FIG. **8**A is a perspective view of an adapter segment;

FIG. **8**B is a cutaway view of the adapter segment;

FIG. **9** is a cutaway view of one or more embodiments having two scalable speakers with batched cones;

FIG. **10** is one or more embodiments of the base segment having a reduced diameter carrier tube, or push-rod;

FIG. **11** is a cutaway view of one or more embodiments of a base segment, two batch segments, and an end segment having a reduced diameter carrier tube, or push rod;

FIG. **12** is cutaway view of an arrangement of carrier tubes, or push rods, from adjacent segments;

FIG. **13** is a cutaway view of an arrangement of carrier tubes, or push rods, from adjacent segments;

FIG. **14** is a cutaway view of an arrangement of carrier tubes, or push rods, from adjacent segments;

FIG. **15** is a cutaway view of one or more embodiments of a batch segment, or end segment, having a spider centering and/or holding the push rod or elongated voice coil; and

FIG. **16** is a cutaway view of one or more embodiments of a batch segment, or end segment, having a surround centering and/or holding the push rod or elongated voice coil.

Elements and steps in the figures are illustrated for simplicity and clarity and have not necessarily been rendered according to any sequence. For example, steps that may be performed concurrently or in different order are illustrated in the figures to help to improve understanding of embodiments of the present disclosure.

DETAILED DESCRIPTION

While various aspects of the present disclosure are described with reference to FIGS. **1-16**, the present disclosure is not limited to such embodiments, and additional modifications, applications, and embodiments may be implemented without departing from the present disclosure. In the figures, reference numbers will be used to illustrate the same components. Those skilled in the art will recognize that the various components set forth herein may be altered without varying from the scope of the present disclosure.

FIG. **1** is a perspective view of one or more embodiments of a scalable speaker with batched cones assembly **100**. The scalable speaker with batched cones assembly **100** has at least one base segment **200**, one or more batch segments **300**, and an end segment **400**. Each speaker with batched cones assembly **100** is scalable through the number and combination of base, batch, and end segments that are used in the assembly. In the embodiment shown in FIG. **1**, two scalable speakers with batched cones assemblies **102** and **104** may be attached to each other with an optional adapter segment **800**.

In-vehicle applications typically have packaging requirements that require an assembly be as small as possible, yet a listener's expectation of bass-performance remains high. The scalable speaker with batched cones assembly **100** of the inventive subject matter presents many more options for integrating the speaker **100** within the vehicle cabin, using little packaging space while maintaining bass-performance that meets, or exceeds, the listener's expectation. For example, one or more scalable speakers with batched cones assemblies **100** may be placed below a rear seat in the vehicle cabin where it uses a cavity below the rear seat as working volume. Using a subwoofer with a conventional speaker in this location is nearly impossible due to subwoofer size. However, the scalable speaker with batched cones assembly can accommodate a plurality of batch segments to provide more cone area. By stacking the radiating area of multiple cones over each other, the assembly maintains an overall effective cone surface for a compact assembly allowing it to be packaged into a smaller space without limiting sensitivity or efficiency. Furthermore, the scalability of the batched segments makes the design customizable across different applications, limited only by the size of the location within which the speaker with batched cones assembly will be installed.

One or more embodiments of a scalable speaker with batched cones assembly **100** is shown in FIG. **1**. The assembly **100** has at least one of first and second subassemblies, i.e., scalable speakers **102**, **104**. In an example arrangement shown in FIG. **1**, two sub-assemblies **102**, **104** with batched cones are arranged facing each other. Each speaker sub-assembly **102**, **104** has an active end with at least one active segment **102*a***, **104*a*** at the base segment **200**, at least one batch segment **300**, and a passive end **102*b***, **104*b*** at the end segment **400**. In one or more embodiments, the first scalable speaker with batched cones **102** is connected at the passive end **102*b*** with the passive end **104*b*** of the second scalable speaker with batched cones **102** by way of the optional adapter segment **800** to configure the speaker as a dual-core speaker with batched cones assembly **100**. However, it should be noted that it is not required that scalable speakers with batched cones **102**, **104** be connected as shown and that one of the first **102** or second **104** scalable speakers with batched cones may be a standalone unit. It should be noted that the speaker with batched cones **102**, **104** unit may have fewer, or more, batch segments than what is shown in FIG. **2**.

It should be noted that assembly **100** may include several scalable speakers with batched cones **102** and **104** arranged in a row within the same enclosure. It should also be noted that the assembly **100** may include more active elements (base segments **200**) stacked on top of each other, resulting in a summing of a motor force of each base segment **200**.

Referring now to FIGS. **2**A and **2**B, the base segment **200** is described. The base segment **200** is an active unit meaning that it is electrically driven by a motor **204**. The motor has a magnet **207** that may be a permanent slug or ring neodymium magnet, or an electromagnet. A perspective view of the base segment **200** is shown in FIG. **2**A. A cross-sectional view of the base segment **200** is shown in FIG. **2**B. Each base segment **200** has a base enclosure **202** that houses a motor assembly **204**, a voice coil **205**, a carrier tube **206** or push rod (hereinafter, carrier tube), a cone **208**, and a spider **210** (or surround). The carrier tube **206** passes from the voice coil **205** through the cone **208** and spider **210** and provides support for the cone **208**. A top end **212** of the carrier tube **206** has a flange **214** for connecting the base segment **200** to a batch segment (not shown in FIG. **2**A or **2**B). The carrier tube **206** is a predetermined length, l, so that the top (or second) end **212** extends beyond the cone **208** and a top side **216** of the base enclosure **202**. The carrier tube **206** may have a flange **214** for connecting to a carrier tube of an adjacent segment (not shown). Further, the carrier tube **206** may be more than one component as shown in FIG. **2**B.

In the example shown in FIGS. **2**A and **2**B, a first side **222** of the cone **208** is coupled to a first resonance volume, such as a vented resonance volume of an adjacent enclosure (i.e., a batch or end segment to be described hereinafter. A second side **218** of the cone **208** couples to a second resonance volume that is different from the first resonance volume, such as a resonance volume of the base enclosure **202**. The second side **218** of the cone **208** is coupled to the second resonance volume by way of an opening **220** in the enclosure **202**. For example, opening **220** may vent into a vehicle cabin. Coupling front and back radiated signals to different resonance volumes, first and second resonance volumes, prevents acoustic short.

Each batch segment **300** is a passive unit of the speaker with batched cones assembly **100**. FIG. **3**A shows a perspective view of the batch segment **300**. FIG. **3**B shows a cross-sectional view of the batch segment. The batch segment **300** has a batch segment enclosure **302** that houses a carrier tube **306**, a cone **308**, and a suspension element **310**, such as a spider or surround. The carrier tube **306** passes through the cone **308** and suspension element **310** and provides support during excursion of the cone **308**. A top end **312** of the carrier tube **306** has a flange **314** for connecting to a carrier tube of another batch segment **300** or an end segment (not shown in FIG. **3**A or **3**B). A bottom end **313** of the carrier tube **306** is configured to attach to either the base segment (not shown in FIG. **3**A or **3**B) or a top end **312** of the carrier tube **306** of another batch segment **300** (also not shown in FIG. **3**A or **3**B). The carrier tube **306** has a predetermined length **1**, so that the top end **312** with the flange **314** extends beyond a top side **316** of the batch segment enclosure **302**.

A first side **322** of the cone **308** in the batch **300** segment is coupled to a first resonance volume by way of the enclosure of an adjacent batch or end segment (not shown). The batch enclosure **302** has an opening **320** that couples a second side **318** of the cone **308** of the batch segment to a second resonance volume that differs from the first resonance volume, such as, for example, the vehicle cabin. The enclosure **302** that attaches to the adjacent batch or end segment (not shown) has an opening **326**. In one example, the opening **326** provides exposure to a second resonance volume for the first surface **222** of the cone **208** in the base segment **200**.

The batch segment **300** is a passive segment. It is mechanically driven by the carrier tube **306** being connected to supporting, and/or centering, the voice coil **208** of the active base segment **200**, which is electrically driven by the magnet and the voice coil as described above with reference to FIG. **2**B.

FIG. **4**A shows a perspective view of an end segment **400**. FIG. **4**B shows a cross-sectional view of the end segment **400**. The end segment **400** has an end segment enclosure **402** that houses a carrier tube **406**, a cone **408**, and a spider **410** (or surround). The carrier tube **406** passes through the cone **408** and spider **410** and provides support for the cone **408**. A bottom end **413** of the carrier tube **406** has a flange **414** for connecting to a carrier tube of a batch segment **300** (not shown in FIGS. **4**A and **4**B). A top end **412** of the carrier tube **406** has a dust cap **424**. The carrier tube **406** has a predetermined length, l, so that the top end **412** with the dust cap **424** extends beyond a top side **416** of the end segment enclosure **402**. A first side **422** of the cone **408** in the end segment **400** is exposed to a resonance volume by way of, for example, an adjacent enclosure (not shown). The batch enclosure **402** has an opening **420** that couples a second side **418** of the cone **408** to a second resonance volume. An opening **426** couples a resonance volume of the first side **222**, **322** of the cone **208**, **308**, of an adjacent segment, either base segment **200** or a batch segment **300**.

As described above, there is no motor assembly in the one or more batch segments **300** or the end segment **400**. Instead, the batch segments **300**, and the end segment **400** are mechanically driven through their connection to the base segment and/or each other. For example, the carrier tube **306** of the batch segments **300** may connect to each other and to the carrier tube **406** in the end segment **400**. Ultimately, the passive segments **300**, **400** are mechanically driven by use of carrier tubes and/or push rods between each passive segment **300**, **400** and the active segment **200** which is electrically driven. This may be done by connecting the cones **308**, **408**, of each batch and end segment to the carrier tubes **306**, **406**.

Alternatively, the batch segment **300** and end segment **400** cones **308**, **408** may not be connected to the carrier tubes **306**, **406**. In this example, the segments **300** and **400** behave as passive radiators whose resonance frequency may be tuned by suspension force and weight to extend low-frequency.

The enclosures **202**, **302**, and **402** may be mechanically fastened to each other. The openings should align. For example, the opening **220** in the base segment enclosure **202** should align with the opening **320** in each of the batch segment enclosures **302**, and the opening **420** in the end segment enclosure **402**.

In one or more embodiments, the speaker with batched cones assembly **500** may include at least one base segment **200** and at least one end segment **400**. One or more batch segments **300** may be connected in between a base unit **200** and the segment **400**. FIG. **5**, for example, shows a cross sectional view of a speaker with batched cones assembly **500** having a base segment **200**, first and second batch segments **300***a*, **300***b* attached thereto, and an end segment **400** attached to one of the batch segments **300***b*. The first batch segment **300***a* is attached to the base segment **200**, The second batch segment **300***b* is attached to the first batch segment **300***a*, and an end-unit batch segment **400***a* is attached to the second batch segment **300***b*. The number, n, of batch segments **300** is scalable **300***a*-**300***n*, and may vary according to packaging constraints, performance considerations, etc. It should be noted that the speaker assembly with batched cones may also have only one base segment **200** and one end unit **400**, without intermediate batch segments **300**.

In the one or more embodiments shown in FIG. **5**, the speaker with batched cones assembly **500** has a carrier tube **206** that is one piece. It should be noted that this is shown for example purposes only, and as described above, it is possible that the carrier tube may be multiple components. In yet another example, the carrier tube may be one or more push rods. The design of the speaker with batched cones assembly having one or more pushrods, or a carrier tube of one or more components may depend upon its specific use in the environment and a packaging location for the assembly.

The speaker with batched cones assembly **500** may be standalone. In one or more embodiments of a standalone speaker with batched cones assembly, a cover segment **600** may be attached to the passive end. FIG. **6**A shows a perspective view of a cover segment **600**. FIG. **6**B shows a cutaway view of the cover segment **600**. The cover segment **600** has a cover enclosure **602** with an opening **626** that couples to a respective resonance volume, for example the first resonance volume for the first side of the cone of the adjacent segment, which may be a batch segment **300** or an end segment **400**, for example.

FIG. **7**A is a perspective view of one or more embodiments of a speaker with batched cones **700** showing one base segment **200**, two batch segments **300***a*, **300***b*, an end segment **400**, and a cover segment **600**. FIG. **7**B is a cross sectional view. A first batch segment **300***a* is attached to the base segment **200**, a second batch segment **300***b* is attached to the first batch segment **300***a*. The end segment **400** is attached to the second batch segment **300***b*. In the example shown in FIG. **7**, the cover segment **600** attaches to the end segment **400**.

As discussed earlier herein, in one or more embodiments, speakers with batched cones **700** may be attached to each other by way of the adapter segment **800** to create a dual-core speaker with batched cones. FIG. **8**A is a perspective view of the adapter segment **800**. FIG. **8**B is a cutaway view of the adapter segment **800**. The adapter segment **800** is an enclosure **802** configured to fasten the end segment of each speaker with batched cones **700** to each other. The adapter **800** has an opening **820** that couples towards the resonant volume and an opening **826** that couples towards a respective resonance volume, either the first or the second resonance volume, to avoid acoustic short circuit.

FIG. **9** is a perspective view of one or more embodiments showing two speaker with batched cones assemblies **500***a* and **500***b* joined by the adapter segment **800** to create a dual-core speaker with batched cones assembly **900**. FIG. **9**B is a cutaway view of the dual-core speaker with batched cones **900**. A passive end **502***b* of the first speaker with batched cones **500***a* is joined with the passive end **504***b* of the second speaker with batched cones **500***b* at the adapter segment **800**.

FIG. **10** is a cutaway view of one or more embodiments of a base segment **1000** having a carrier tube **1002**, with a diameter that, ideally but not limited to, may be smaller than or equal to a diameter **1006** of the voice coil **1003** driving the active unit. For example, a first end **1004** of the carrier tube **1002** has a first diameter **1006** that is approximately as large as the diameter **1006** of the voice coil **1003** to ensure high performance of the motor. A second end **1008** of the carrier tube **1002** has a second diameter **1010** that may be smaller than the first diameter **1006** to allow a larger cone surface contributing to the effective overall cone area of the whole assembly.

FIG. **11** is a cutaway view of one or more embodiments of a speaker with batched cones **1100***a* having the reduced diameter carrier tube **1102** at a base segment **1104** discussed in FIG. **10**. Each batch segment **1106**, **1108** has a carrier tube **1110** with a diameter that matches the reduced diameter of the carrier tube **1102** of the base segment. The end unit **1112** has a carrier tube **1114** with a diameter that matches the diameter of the carrier tube **1110** of the batch segments **1106**, **1108**. Inner surrounds **1116** may be added to the carrier tube of the batch segments **1106** to provide more overall effective cone surface.

In one or more embodiments, the speaker with batched cones does not have additional push rod or carrier tube elements. See for example the cutaway view shown in FIG. **7**B. An extended voice coil in the base segment **200** upon which additional batched cones, surrounds or spiders may be attached directly. It should be noted that a combination is also possible in which a single push rod, or carrier tube, extends from the voice coil in the base segment **200** through the additional segments, batched cones, surrounds or spiders.

For one or more embodiments that have carrier tubes having multiple components, there are several alternatives for connecting the carrier tube components. For example, FIG. **12** shows a cutaway exploded view of one or more embodiments of an arrangement **1200** for connecting carrier tubes **1202***a*, **1202***b* between segments. A mating end **1204***a* of a first carrier tube **1202***a* has a flange **1206***a* that extends perpendicular to the carrier tube around an outer circumference of the carrier tube **1202***a*. A mating end **1204***b* of a second carrier tube **1202***b* has a flange **1206***b* that extends perpendicular to the second carrier tube **1202***b* around an outer circumference of the second carrier tube **126***b* such that the mating ends **1204***a*, **1204***b* of the carrier tubes **1202***a*, **1202***b* abut each other when assembled and meet circumferentially at the flanges **1206***a*, **1206***b*. The mating ends may be glued at the flanges **1206***a*, **1206***b*.

FIG. **13** shows a cutaway exploded view of one or more embodiments of an arrangement **1300** for connecting carrier tubes **1302***a*, **1302***b* between segments. A mating end **1304***a* of the carrier tube **1302***a* has a circumferential collar **1306**. The collar **1306** has an inner diameter that receives an outer diameter of a mating end **1304***b* of the carrier tube **1302***b*. The connection may be reinforced by gluing the carrier tubes **1302***a*, **1302***b*, at the connection.

FIG. **14** shows a cutaway exploded view of one or more embodiments of an arrangement **1400** for connecting carrier tubes **1402***a*, **1402***b* between segments at mating ends **1404***a* and **1404***b* of the carrier tubes **1402***a*, **1402***b*. The mating ends **1404** of each carrier tube **1402***a*, **1402***b* abut each other coaxially, and may be glued. A support ring **1406** is externally coupled to an outer diameter of each of the carrier tubes **1402***a*, **1402***b*. The support ring **1406** may be glued to an outer perimeter of the carrier tubes at their respective mating ends **1404***a*, **1404***b*.

It should be noted that the batch segments may have a spider instead of a surround as shown in FIG. **15**. FIG. **15** shows a cutaway view of a portion of a base segment **1502** and a batch segment **1504** of a speaker with batched cones assembly **1500**. The batch segment **1504** has a spider **1506** centered by the carrier tube and affixed to the batch segment enclosure **1506**. A surround **1508** is at the top of the batch segment enclosure.

FIG. **16** show a cutaway view of a base segment **1602** and a batch segment **1604** of a speaker with batched cones assembly **1600**. The batch segment **1604** has a surround **1606** affixed to the bottom of the enclosure and a surround **1608** at the top of the enclosure. In the embodiment shown in FIG. **16**, the surround **1606** is in an opposite direction of

the surround **1608**. It should be noted that it is possible for both the surround **1606** and **1608** to have the same direction. The surround geometry serves to separate the back and front sides of the radiating area to avoid acoustic short and air vent noise. The surround may also serve as an additional guiding feature to control the direction of the carrier tube/push rod. The geometry of the surround described herein is for example purposes and it should be noted that other elements may also achieve acoustic separation. For example, a foam surrounding.

The speaker with batched cones assembly of the inventive subject matter is very easily scalable based on performance requirements, packaging requirements, while maintaining small packaging.

In the foregoing specification, the present disclosure has been described with reference to specific exemplary embodiments. The specification and figures are illustrative, rather than restrictive, and modifications are intended to be included within the scope of the present disclosure. Accordingly, the scope of the present disclosure should be determined by the claims and their legal equivalents rather than by merely the examples described.

For example, the steps recited in any method or process claims may be executed in any order, may be executed repeatedly, and are not limited to the specific order presented in the claims. Additionally, the components and/or elements recited in any apparatus claims may be assembled or otherwise operationally configured in a variety of permutations and are accordingly not limited to the specific configuration recited in the claims. Any method or process described may be carried out by executing instructions with one or more devices, such as a processor or controller, memory (including non-transitory), sensors, network interfaces, antennas, switches, actuators to name just a few examples.

Benefits, other advantages, and solutions to problems have been described above about particular embodiments; however, any benefit, advantage, solution to problem or any element that may cause any particular benefit, advantage or solution to occur or to become more pronounced are not to be construed as critical, required or essential features or components of any or all the claims.

The terms "comprise," "comprises", "comprising", "having", "including", "includes" or any variation thereof, are intended to reference a non-exclusive inclusion, such that a process, method, article, composition, or apparatus that comprises a list of elements does not include only those elements recited but may also include other elements not expressly listed or inherent to such process, method, article, composition, or apparatus. Other combinations and/or modifications of the above-described structures, arrangements, applications, proportions, elements, materials, or components used in the practice of the present disclosure, in addition to those not specifically recited, may be varied, or otherwise particularly adapted to specific environments, manufacturing specifications, design parameters or other operating requirements without departing from the general principles of the same.

What is claimed is:

1. A speaker assembly, comprising:

a base segment enclosure having a motor, a voice coil, and a carrier tube, a first end of the carrier tube rests above the motor, the carrier tube passes through and supports the voice coil, the base segment enclosure has a first opening in a first direction toward a resonant volume and a second opening in a second direction different than the first direction;

at least one batch segment enclosure having a cone and a carrier tube, a first end of the carrier tube of an adjacent batch segment enclosure passes through the cone and aligns with the second carrier tube of the base segment enclosure, each batch segment enclosure has a first opening in the first direction and a second opening in the second direction, wherein the first opening in the base segment enclosure aligns with the first opening in each batch segment enclosure and the second opening in the base segment enclosure aligns with the second opening in each batch segment enclosure;

an end segment enclosure having a cone and a carrier tube, the carrier tube of the end segment enclosure passes through and centers the cone, a first end of the carrier tube of the end segment enclosure aligns with a second end of the carrier tube of the at least one batch segment enclosure, a dust cap is attached to a second end of the carrier tube of the end segment;

the base segment enclosure connects to the at least one batch segment enclosure; and

the at least one batch segment enclosure connects to the end segment enclosure.

2. The assembly as claimed in claim **1**, further comprising a plurality of batch segment enclosures arranged between the base segment enclosure and the end segment enclosure, the first end of the carrier tube of a first batch segment enclosure in the at least one batch segment enclosures aligns with the second end of the carrier tube of the base segment enclosure, the second end of the carrier tube of a last batch segment enclosure in the plurality of batch segment enclosures aligns with the first end of the carrier tube of the end segment enclosure, thereby defining a first speaker with batched cones assembly.

3. The assembly as claimed in claim **2**, wherein a first speaker with batched cone assembly is aligned in parallel adjacent to at least one additional first speaker with batched cone assembly.

4. The assembly as claimed in claim **2**, further comprising at least a second speaker with batched cones assembly having a predetermined number of batch segment enclosures arranged between the base segment enclosure and the end segment enclosure, the first end of the carrier tube of a first batch segment enclosure in the plurality of batch segment enclosures aligns with the second end of the carrier tube of the base segment enclosure, the second end of the carrier tube of a last batch segment enclosure in the plurality of batch segment enclosures aligns with the first end of the carrier tube of the end segment enclosure; and

an adapter segment joining the end segment enclosure of the first speaker with batched cones assembly to the end segment enclosure of the second speaker with batched cones assembly thereby defining a dual-core speaker with batched cones.

5. The assembly as claimed in claim **4**, further comprising a plurality of dual-core speakers with batched cones arranged in parallel.

6. The assembly as claimed in claim **4**, further comprising a plurality of dual-core speakers with batched cones arranged linearly.

7. The assembly as claimed in claim **1**, wherein the carrier tube of the base segment enclosure has a first diameter at the first end and a second diameter at the second end, the second diameter is less than the first diameter.

8. The assembly as claimed in claim **1**, wherein the carrier tube of the base segment enclosure, the carrier tube of the batch segment enclosure, and the carrier tube of the end segment enclosure pass through a center of the base segment enclosure, a center of the at least one batch segment enclosure, and a center of the end segment enclosure; and

the cone in each batch segment enclosure is connected to the carrier tube in each batch segment enclosure.

9. The assembly as claimed in claim **8**, further comprising:

at least a second speaker with batched cones assembly having a predetermined number of batch segment enclosures arranged between the base segment enclosure and the end segment enclosure, the first end of the carrier tube of a first batch segment enclosure in the plurality of batch segment enclosures aligns with the second end of the carrier tube of the base segment enclosure, the second end of the carrier tube of a last batch segment enclosure in the plurality of batch segment enclosures aligns with the first end of the carrier tube of the end segment enclosure;

an adapter segment joining the end segment enclosure of the first speaker with batched cones assembly to the end segment enclosure of the second speaker with batched cones assembly thereby defining a dual-core speaker with batched cones.

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